

Title

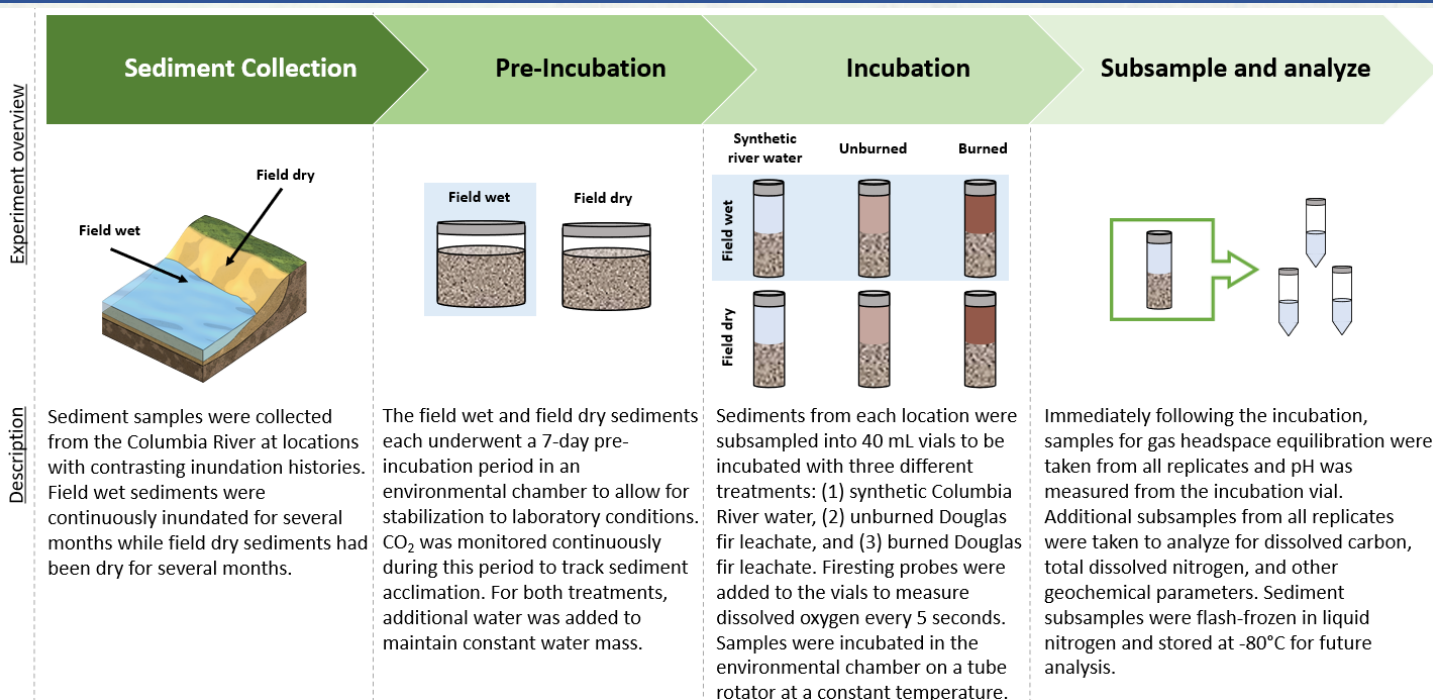
Pyrogenic Organic Matter Laboratory Experiment: Aerobic Respiration and Geochemistry from Variably Inundated Stream Sediments

Summary

This dataset supports a broader study examining the effects of variable inundation and pyrogenic organic matter on ecosystem respiration. The dataset provides data generated from a laboratory batch experiment investigating the interaction between variable inundation conditions (wet and dry sediment) and pyrogenic organic matter (burned and unburned treatments). The contents include time series dissolved oxygen, sediment geochemistry data, and field metadata (including qualitative information on instream and river corridor characteristics). Additional data types, including respiration, will be added as they become available.

For details on how to navigate data packages generated by this project, see <https://data.ess-dive.lbl.gov/portals/PNNLRiverCorridorSFA/About>.

Experimental Design



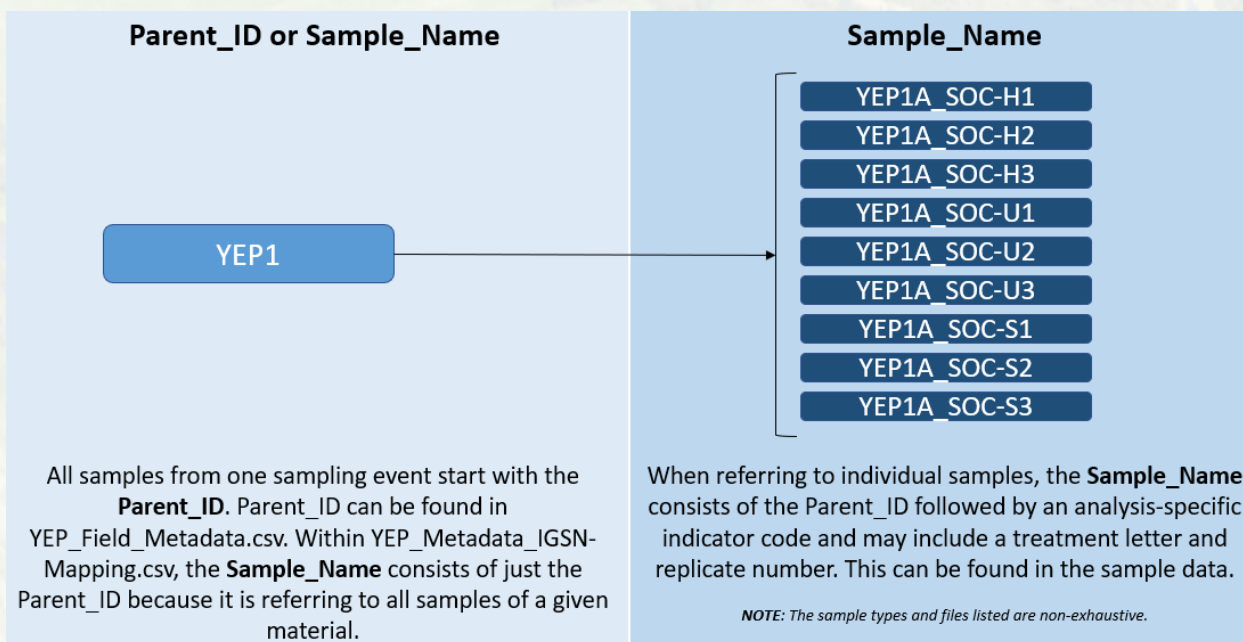
Brief Overview of Methods

Sediment samples were collected from the Columbia River at locations with contrasting inundation conditions – field wet sediments that had been continuously inundated for several months and field dry sediments from a higher elevation on the same shoreline that had been dry for several months. Site metadata, including general environmental information, were manually recorded in the field. See YEP_Sampling_Protocol.pdf for details on sample collection and site metadata collection. Briefly, sediment samples were sieved to less than 2 millimeters and transported on blue ice to Pacific Northwest National Laboratory where they underwent a 7-day pre-incubation period in an environmental chamber to allow stabilization to laboratory conditions. After the pre-incubation period,

sediment incubations were conducted using three treatments: (1) synthetic Columbia River water (0 mg C/L) as a negative control, (2) unburned Douglas fir leachate at 9 mg C/L, and (3) burned (high severity) Douglas fir leachate at 9 mg C/L. All treatments were applied to both field wet and field dry sediments. Aerobic respiration rates were calculated from the slope of the linear regression between dissolved oxygen data measured using FireSting oxygen probes and incubation time. Incubations were monitored continuously with measurements every 5 seconds, with experiments ending after 6 hours or when dissolved oxygen levels reached 2.0 mg/L. For details regarding laboratory and instrument methods, see the alphanumeric methods codes located in the header rows of the chemistry data csv file and their associated definitions in YEP_Methods_Codes.csv. Data formats follow ESS-DIVE Reporting Formats.

Critical Details

- 1 – **Sample Name Mapping:** Each sampling event has a unique Parent_ID in the format YEP#. The field metadata and data files all contain these unique IDs and can be mapped across each other accordingly. The Parent_ID may have other indicators appended when referring to samples. See figure below for details.



- 2 – **Sample Name:** The sample names contain pertinent information for understanding the data (i.e., material type, treatment, replicate), see “YEP_Sample_Name_Metadata.csv” to understand the details of each sample name. *Note: the sample names in the file may go beyond what is published in this data package.*
- 3 – **Replicate Merging:** Each sample name ends with a replicate number. Each replicate indicates an individual reactor. If you would like to merge the data files across different data types, you can do that on the basis of the replicate number (given that the sample name is the same, except the analyte code). For example the row of data for YEP1A_GAS_S1 can be merged with the row of data for YEP1A_INC_S1 because the names are identical aside from GAS and INC, which are the analyte codes.

- 4 – **Replicate Numbers:** Replicates were incubated on different days after field sample collection. In most cases, replicates 1 and 2 were incubated on Day 1, replicate 3 was incubated on Day 2, and replicates 4 and 5 were incubated on Day 3. However, the burned replicate 1 (-H1) and the unburned replicate 2 (-U2) were incubated on Day 2.
- 5 – The leachates used for the treatments were created using leftover material from the BSLE_0013 and BSLE_0050 chars (Grieger et al., 2022). Metadata about the chars and methods used to create the leachates can be found at <https://data.ess-dive.lbl.gov/view/doi%3A10.15485%2F1894135> (Grieger et al., 2022).
- 6 – Partial pressure of CO₂ was converted to molar units using the temperature-dependent Henry's Law constant (K_H) as seen in Regier et al. (2023):

$$K_H = e(A1+A2*(100/T)+A3*\ln(T/100)+S*(B1+B2*(T/100)+B3*(T/100)^2))$$

where A1-3 and B1-3 are gas specific coefficients for the temperature (T) and salinity (S) dependence of solubilities. All incubations were conducted at 21°C, with an estimated salinity of 0.1 PSU. K_H was multiplied by the partial pressure of the gas (p) to get the dissolved molar concentration (C):

$$C = K_H * p$$

Data Package Structure

In addition to this readme, this data package also includes a file-level metadata (FLMD) file that describes each file and a data dictionary (DD) that describes all column/row headers and variable definitions.

This dataset is comprised of (1) file-level metadata; (2) data dictionary; (3) field metadata; (4) international generic sample number (IGSN) mapping file; (5) readme; (6) field protocol; (7) sample name metadata; (8) an environmental context picture for the dry and inundated sampling locations; and (9) a subfolder with sample data from the sediment incubation experiment. The sample data subfolder contains (1) dissolved organic carbon (DOC, measured as non-purgeable organic carbon, NPOC); (2) total nitrogen (TN); (3) gravimetric moisture; (4) carbon dioxide; (5) field wet sediment mass, dry sediment mass, water mass, and field wet sediment volume in incubation and sediment NPOC/TN vials; (6) methods codes; and (7) raw time series dissolved oxygen and temperature. All files are .csv or .pdf.

Acknowledgements, Citations, and License

Acknowledgments

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Citations

- Grieger S ; Aronstein P ; Bailey J ; Barnes M ; Barton R ; Bladon K D ; Chu R ; Forbes B ; Garayburu-Caruso V A ; Graham E B ; Goldman A E ; Homolka K ; Kew W ; Lipton A S ; McKeever S A ; Munson K

M ; Myers C R ; Nieto-Pereira N ; O'Day P ; Otenburg O ; Regier T ; Renteria L ; Roebuck A ; Scheibe T D ; Torgeson J M ; Toyoda J G ; Wagner S ; Winston I ; Young R P ; Myers-Pigg A (2022): Organic matter concentration and composition of experimentally burned open air and muffle furnace vegetation chars across differing burn severity and feedstock types from Pacific Northwest, USA (v4). River Corridor and Watershed Biogeochemistry SFA, ESS-DIVE repository. Dataset. [doi:10.15485/1894135](https://doi.org/10.15485/1894135)

- Regier, P., Ward, N. D., Izquierdo, A., Baldwin, A. H., Day, D., McElhinny, J., Patel, K., Vargas, R., Zheng, J., Exchange Consortium, & Myers-Pigg, A. (2023). Coastal inundation regime moderates the short-term effects of sediment and soil additions on seawater oxygen and greenhouse gas dynamics: A microcosm experiment. *Frontiers in Marine Science*, 10, 1308590. <https://doi.org/10.3389/fmars.2023.1308590>

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Change History

Approach to change history and versioning:

Updates to **data package** version: When any file within a data package is updated, the data package version number is updated. The data package version number is indicated in the title of the data package, the data package folder name, and in the change history table below. You can access previous versions of the data package by sending a request to ESS-DIVE.

Updates to **individual file** versions: As files are changed, the file version number is also updated. The file version number is indicated in the file name, file level metadata (flmd) file, and the change history table below. The version number on an individual file may not match the version number of the data package. For example, v3 of a data package may include v2 of an individual file.

Change history:

Data Package Version	Changes
Version 1 <i>November 2025</i>	Original data package publication